

Digging into the Daily Pennsylvanian



Introduction

The Daily Pennsylvanian (fondly referred to as “the DP”) is the newspaper of record for the University of Pennsylvania. It has operated since 1885 and is an independent, entirely student-run newspaper. As three students at Penn we thought it would be an ideal source of text data to apply what we have learned this semester. Fortunately for us, the DP was early to the World Wide Web and so has an article archive stretching back to 1993 with over 86,000 articles to explore. Besides the text of the articles we also have access to engagement metrics for more recent articles allowing us to draw links between article features and engagement. Topic modeling would help us to see what the DP has historically written about and what topics have gained prominence more recently. Using NLP tools covered in class like sentiment analysis we wanted to identify what articles and what topics generate high sentiment and how sentiment about DP articles has changed since 1993.

In the course of our analysis we were able to find answers to most of our questions, some more satisfying than others. Before you read on we’d like to thank the DP for providing access to this data without which this project would not have been possible.

EDA - General

With over 86,000 articles each a few hundred words long, our dataset certainly falls under the realm of “big data”. To get a handle on what we’re looking at let’s first do some general exploratory data analysis.

Most Read Articles

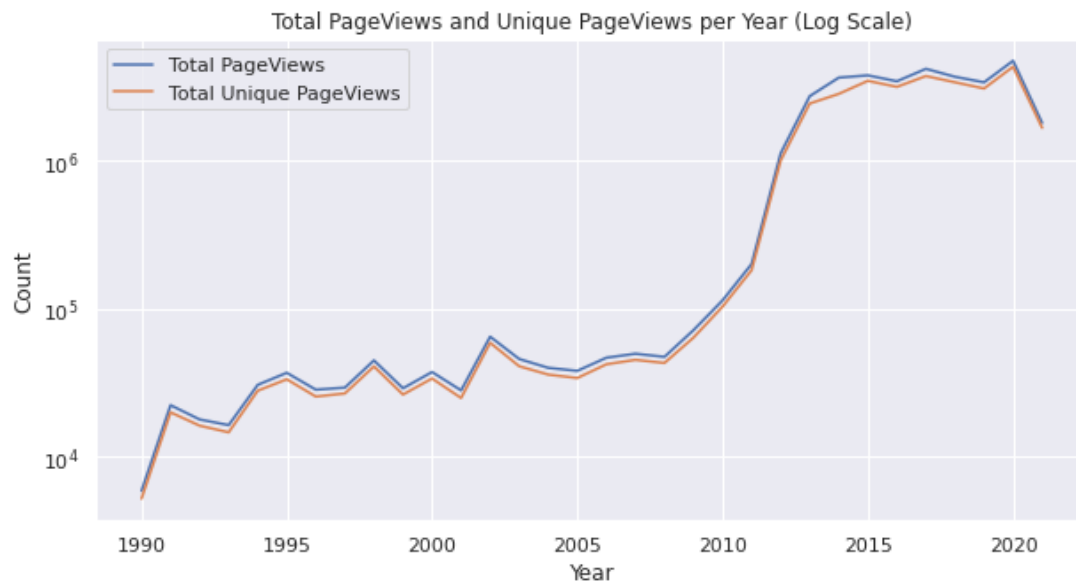
The top ten most read articles in the dataset were:

1. "Was Trump really a top student at Wharton? His classmates say not so much" - 567k views
2. "Trump flaunts Wharton degree, but his college years remain a mystery" - 521k views
3. "U. of Oklahoma student to be suspended for racist group message after Penn Police, FBI investigation" - 276k views
4. "Mapping U.S. colleges' fall 2020 plans" - 142k views
5. "Phi Delta Theta holiday photo sparks controversy" - 129k views
6. "Penn basketball alum Tyrone Gilliams sentenced to 10 years in prison" - 99k views
7. "JOKE ISSUE: Emma Watson chooses Penn for graduate school" - 94k views
8. "CAPS Executive Director Gregory Eells died by suicide Monday morning" - 93k views
9. "Agatha Advincula | I'm a legacy student, and I'm not ashamed" - 92k views
10. "James Fisher | 'Privilege' does not exist to White Penn professors — and they keep 'trying it'" - 91k views

Perhaps unsurprisingly former president Trump is atop this ranking claiming not just the first spot but also the second spot with over half a million views each. Trump's name recognition and lightning pole status no doubt gave these articles a more national spotlight. After the top two articles the pageviews drop substantially but remain high. The bottom five articles are clustered together in view count suggesting the high five figures is a more natural ceiling for the DP's reach when not Trump-focused. Given that any time the Penn student body numbers around 20,000 plus around 5,000 academic staff, reaching an audience of nearly six figures is an impressive feat.

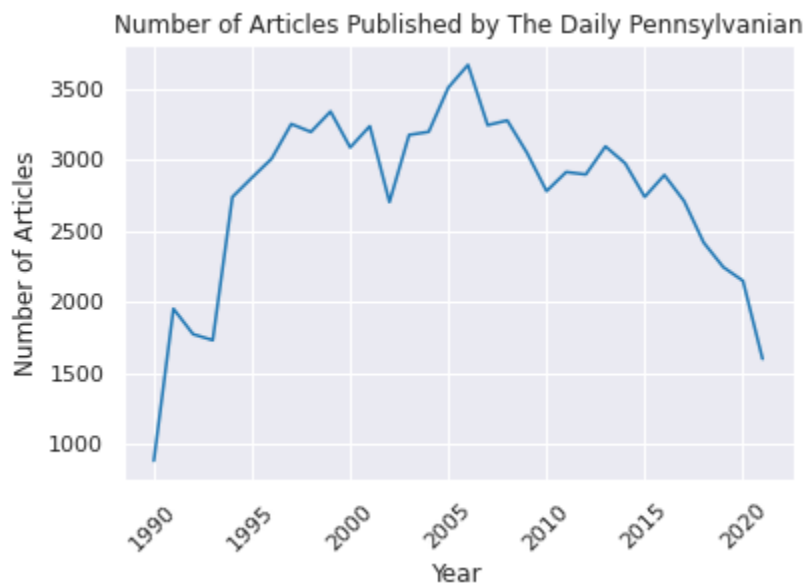
Unlike household newspapers like the New York Times or Wall Street Journal where opinion pieces tend to top the view counts, here opinion pieces only reach 9th and 10th. Also notable is the presence of a humor piece at #7. An important caveat for this analysis is that pageviews naturally weigh more towards more recent articles since the DP only started tracking engagement data through google analytics in the mid 2010s.

From Print to Online



This graph helps visualize the caveat mentioned above as we can see that online page views at the DP were fairly low (note the log scale) until an inflection point around 2011 when they increased by over an order of magnitude. More electronic devices, internet connectivity and an internal strategic focus all likely contributed to this shift towards a more web-based DP.

Pace of Publishing

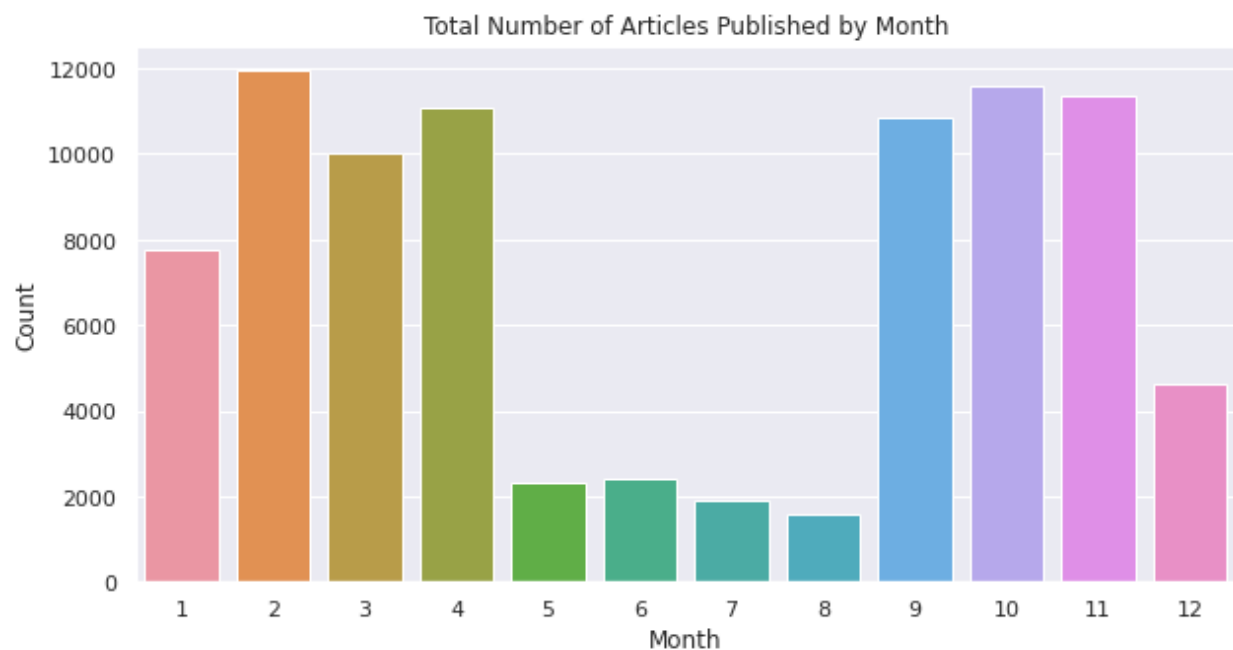


The graph above shows how the number of articles published online by the DP has changed in the past 30 years. As expected there is a rising trend in the first half of the period with the annual articles published rising from below 1000 to peaking at over 3500 in 2006. Our

hypothesis would have been that the entire graph would maintain this upward trend but instead the number of articles published has declined to around 1500 in 2020. The simplest explanation is that perhaps the DP has produced less content in recent years, and this was definitely true in 2020 due to the pandemic. However this seems unlikely given the DP's [recent track record of success](#). An alternative explanation is that the DP's lifestyle and humor publications, 34th Street and Under the Button were shifted to their own sites and so their pieces are not registered in our counts later in the period.

Patterns of Publishing

To conclude our general EDA we look at when the DP tends to publish during the academic year (numbers = months of the year):



Publishing peaks in February with the lightest period during the summer break from May-August and then a consistent high level during the fall semester. February and October have the highest number of articles likely because that's when campus is most active but also when new DP journalists are fully engaged and churning out articles.

Topic Modeling

As one component of the modeling section of our project we decided to use topic modeling, a form of unsupervised machine learning, to see what an untrained but unbiased computer thinks the DP writes about. The specific method we used, Structural Topic Modeling (STM), leverages the content available in each article and the metadata associated with each article to create soft clusterings of articles into topics. The output of the STM is a set of keywords per topic, and a set of documents per topic. Once we had the keywords associated with each topic we used our judgment and contextual knowledge to label the topics. For example one cluster of keywords was “‘club’, ‘sac’, ‘budget’, ‘funding’ ‘society’”, which were grouped together as Clubs.

The Topics

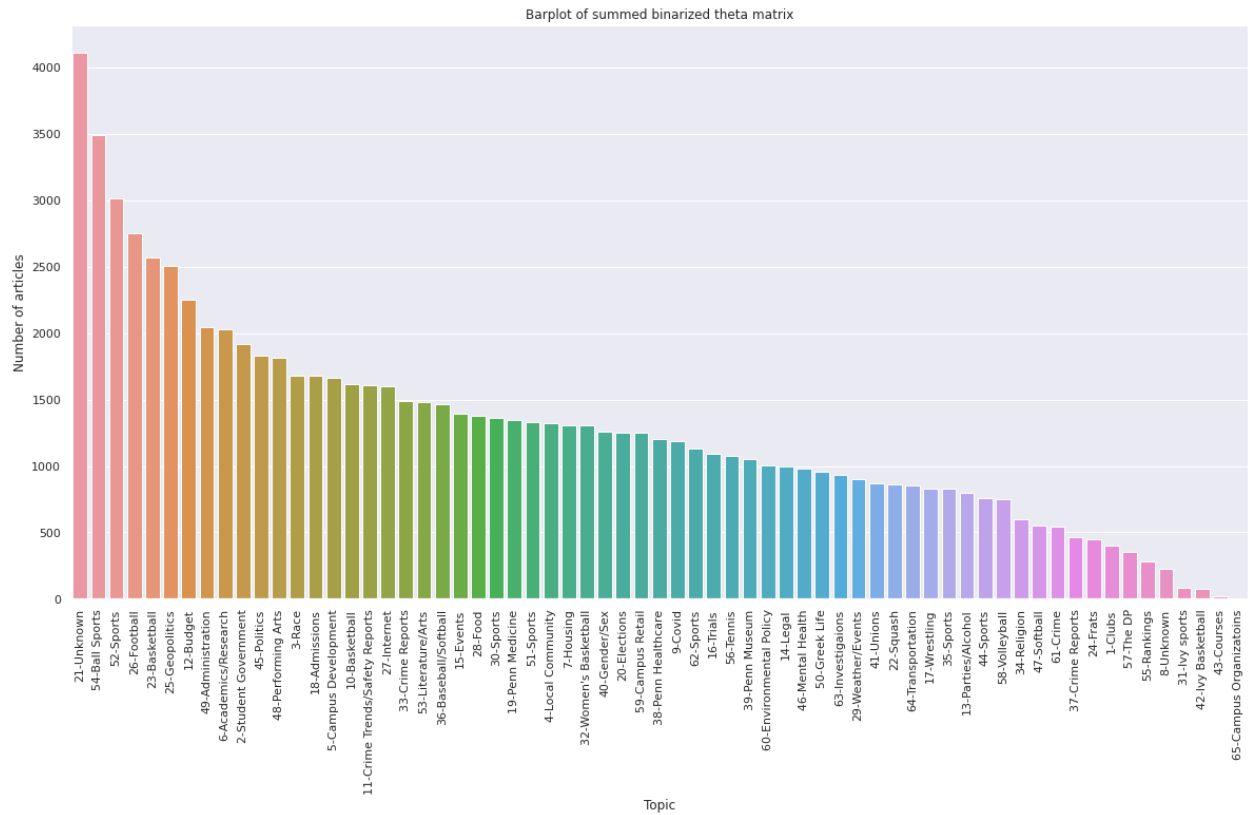
The topics noticeably differ in how broad or niche they are. For example “Campus Retail” is a lot more specific than “Events”. This is perhaps a consequence of how we selected the “K” parameter of the STM which controls the number of topics it generates. We selected a K of 65 since this was what was suggested by the STM package in order to lower overall perplexity which is a measure of how good the topic clusters are. It is entirely possible that with a higher K value, topics such as “Events” could be broken down further into topics like “Protests” and “Celebrations”. Some clusters were too difficult to separately interpret leading to repetition of topics like “Sports” whereas a few were simply not interpretable leading to “Unknown”.

We note that some topics, such as crime appear multiple times. One potential reason for this is that the model could be getting confused by varying writing styles and formats. Since the model has no underlying “understanding” of what constitutes a topic and only uses the co-occurrences of words to form clusters, it’s possible that it may end up forming two separate clusters for crime if one cluster’s language is distinct from another cluster’s.

The word clouds below illustrate what the top words in each topic were.

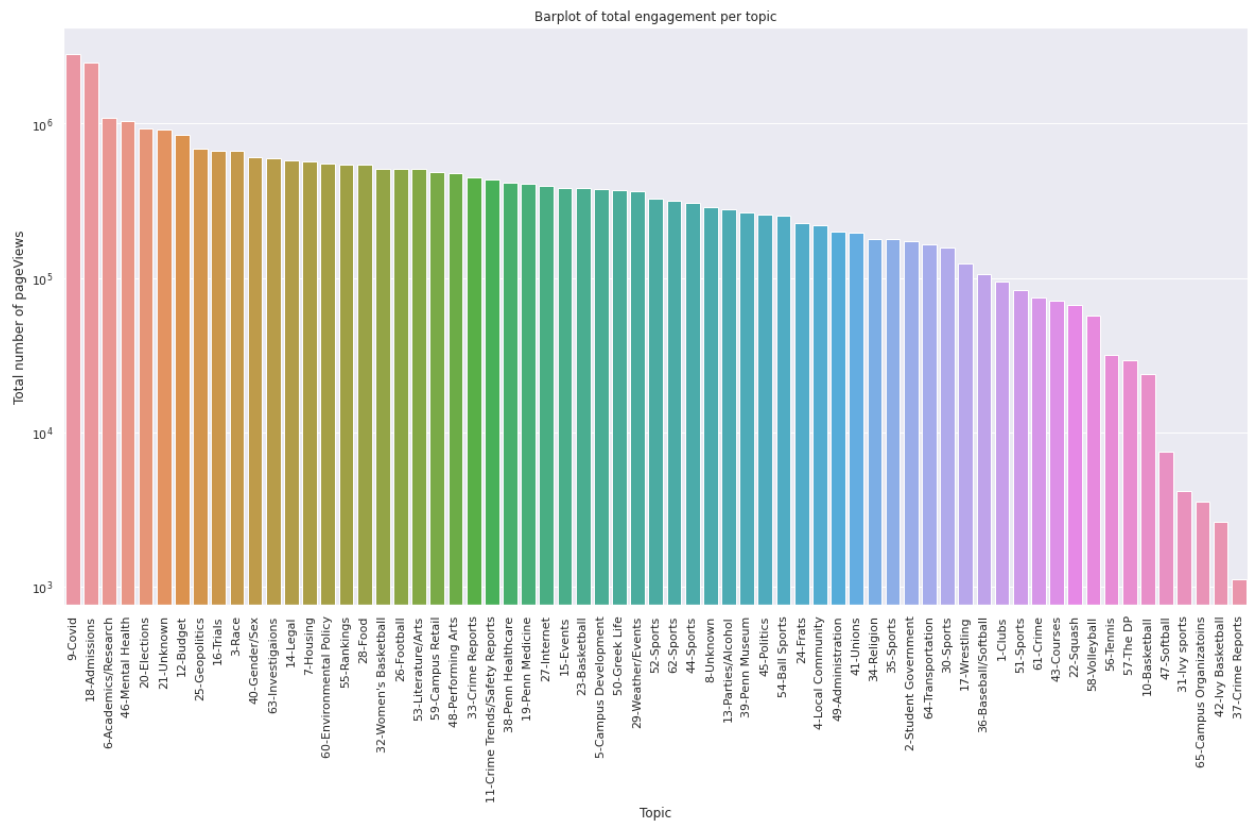


Articles by Topic



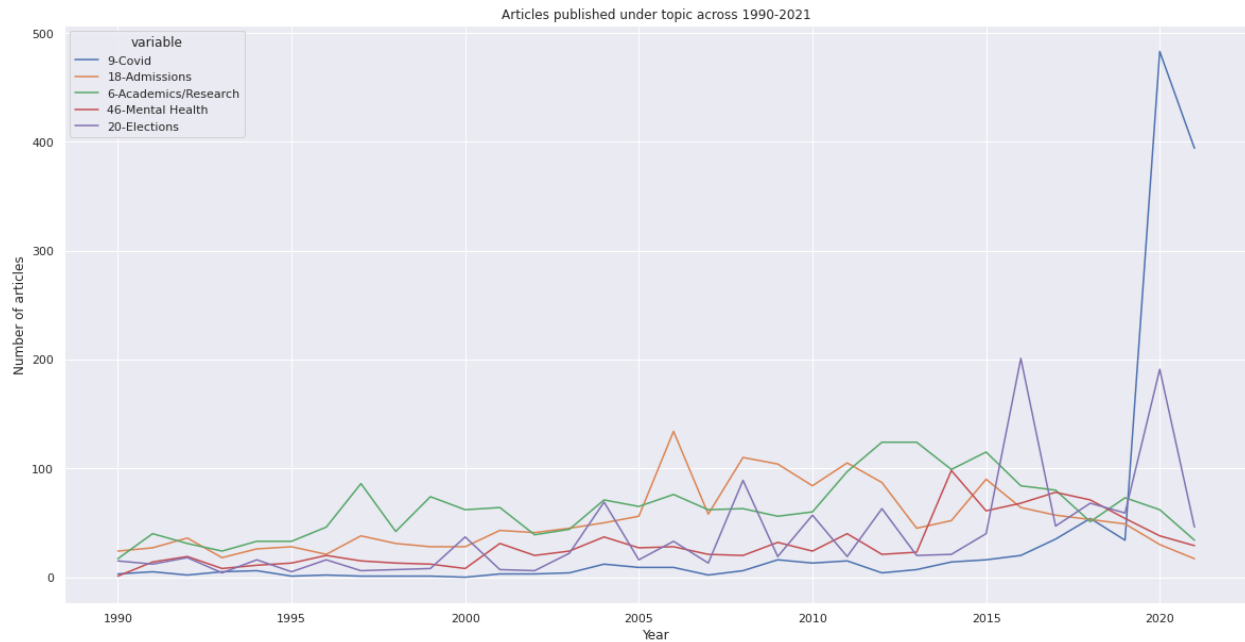
Clearly there is significant variation in how much topics are covered by the DP. Somewhat unhelpfully for us, “Unknown” appears as the most frequently covered topic. Nevertheless it is noteworthy that sports make up four of the top five topics identified with over 2,500 articles on these topics. In contrast, topics like Clubs, Rankings and Crime are on the other end of the spectrum with relatively little coverage. The key takeaway though is the middle of the graph showing the most topics we identified receive roughly equal coverage with somewhere around 1000 articles per topic.

Pageviews by Topic



The graph above shows the pageviews by topic (note the log scale). Here we see more hot-button issues at the top end of the spectrum such as Covid, Admissions and Mental Health. In contrast sports-related topics do not crack the top 10 topics by pageviews and are found more frequently towards the bottom end. One key note in interpreting this graph however is that some topics may contain articles which are predominantly dated (and thus don't contain good engagement tracking). This means that the results of this graph are shaped not only by how "popular" a topic is, but also how recent it is.

Timely Topics



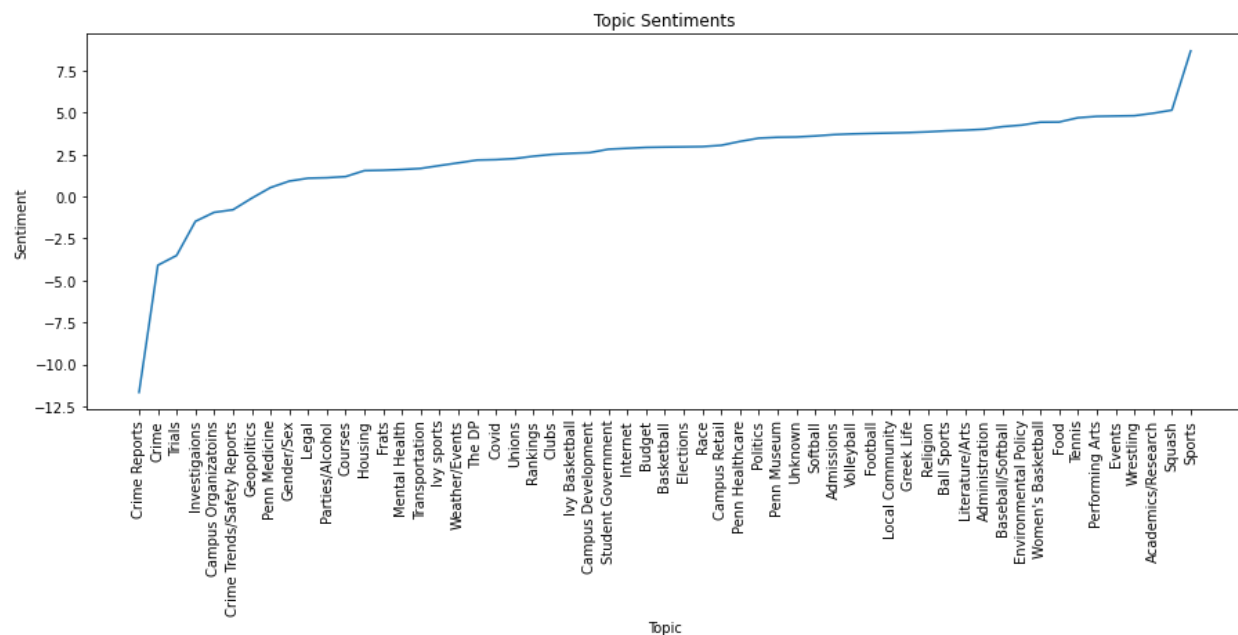
Zooming in on the top five topics by pageviews to see how they've been covered over time reveals that several topics are both highly viewed and timely. Covid in particular stands out for its rapid rise in coverage last year, continuing into this year. As a former DP journalist who covered Penn's rapid shutdown and [transition to online classes last year](#) it was personally interesting to see this coverage spike visualized. Elections also show a spike every four years, in line with presidential elections, particularly the ones involving Donald Trump.

EDA - Sentiment

In the past few years, more attention has been placed on the role traditional media plays in creating narratives. Concerns about the power of media in shaping people's perceptions are well supported by studies on phenomena like [emotional contagion](#), an effect studied by Facebook when they altered the sentiment that appeared on people's feeds. They found that a social media user's emotions can be influenced by the quantity of positive/negative posts they come across. Given the ability of articles to influence emotions, we thought it would be interesting to provide a method to rate article sentiment.

For each article in the DP dataset, we ran a bag-of-words sentiment analyzer (called VADER) to determine the overall positivity, negativity, neutrality, and compound score. We then averaged each score over all articles in each topic to find the average sentiment per topic. Note that the compound score is an aggregate score based on both the positive, negative, and neutral ones which allows us to rank each topic relative to all others. This could be used to tailor social media users' feeds such that each feed has an average sentiment of neutral, eliminating the emotional contagion mentioned above.

Sorted Topic Sentiment

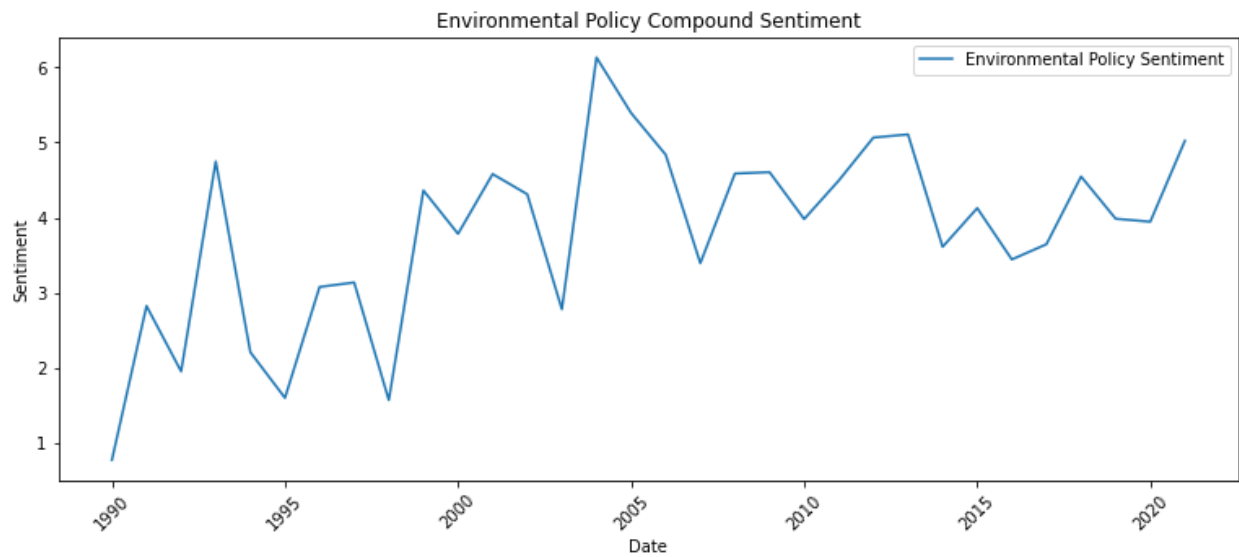


This graph was created by averaging the overall sentiment in each topic cluster. The results of this graph probably seem very well-aligned with our intuitions. Topics such as sports which are more light-hearted sit at the right end with high average sentiment while topics such as crime and geopolitics sit at the left end with low sentiment scores.

With both the sentiment scores and article publication dates for each topic, we are also able to plot the average sentiment of each topic over time. This is another interesting piece of EDA as

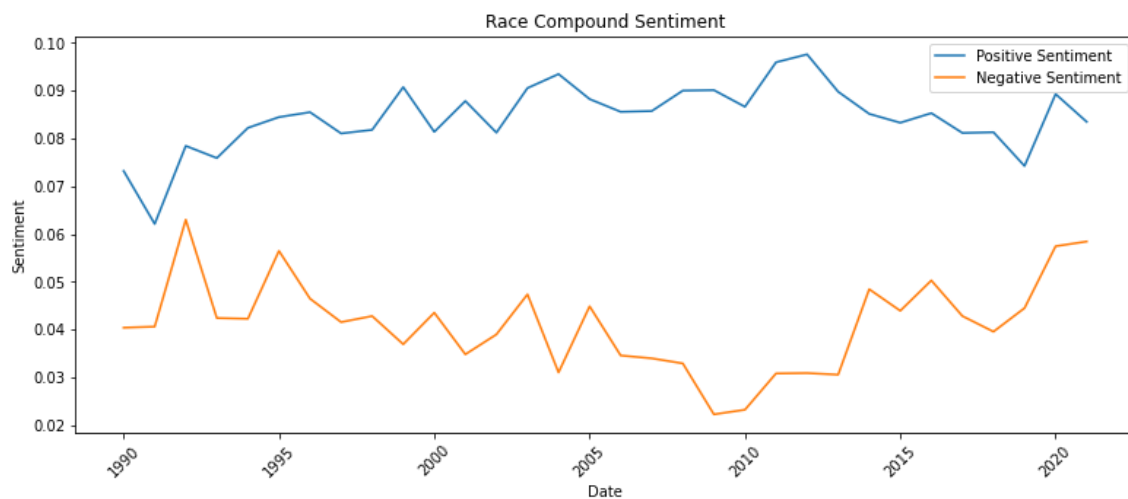
we can see the tone for a given topic varies over time. A couple examples of topic sentiment over time graphs are that of Environmental Policy and Race:

Environmental Policy Sentiment



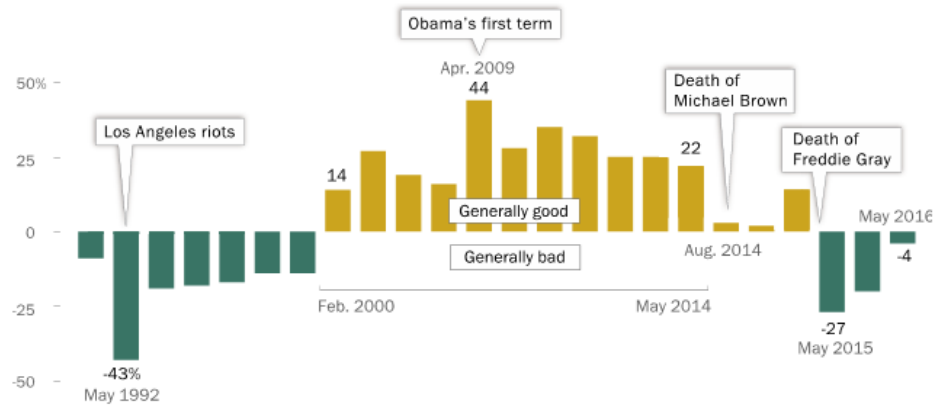
Our graph of Environmental Policy sentiment over time corroborates the push in the 21st century towards moving to sustainable consumption of the earth's resources.

Race Sentiment



Views of race relations are more negative now than they have been for much of the 2000s

Percentage point difference between all adults saying race relations are "generally good" and those saying "generally bad"



Note: Trend data from CBS News and CBS News/New York Times surveys.
Source: Survey of U.S. adults conducted Feb. 29-May 8, 2016, Q5F1.
"On Views of Race and Inequality, Blacks and Whites are Worlds Apart"

PEW RESEARCH CENTER

Our positivity and negativity plots of the Race topic also correlate with the observed race relations found by Pew Research. We note the spike in negative sentiment in 2015 and in 1992 which lines up with historic events.

From the couple examples seen above, our EDA establishes the link between article sentiment and environmental factors such as the outlook on Environmental Policy and Race politics. Thus, we are left with a situation like the infamous chicken or egg dilemma: do article sentiments influence the environment (by emotional contagion), or does the environment cause a change in article sentiment.

Regression

We wanted to predict the engagement of articles with two aims:

1. Analyze which topics tended to have high engagement
2. Figure out how well we can predict engagement in and of itself

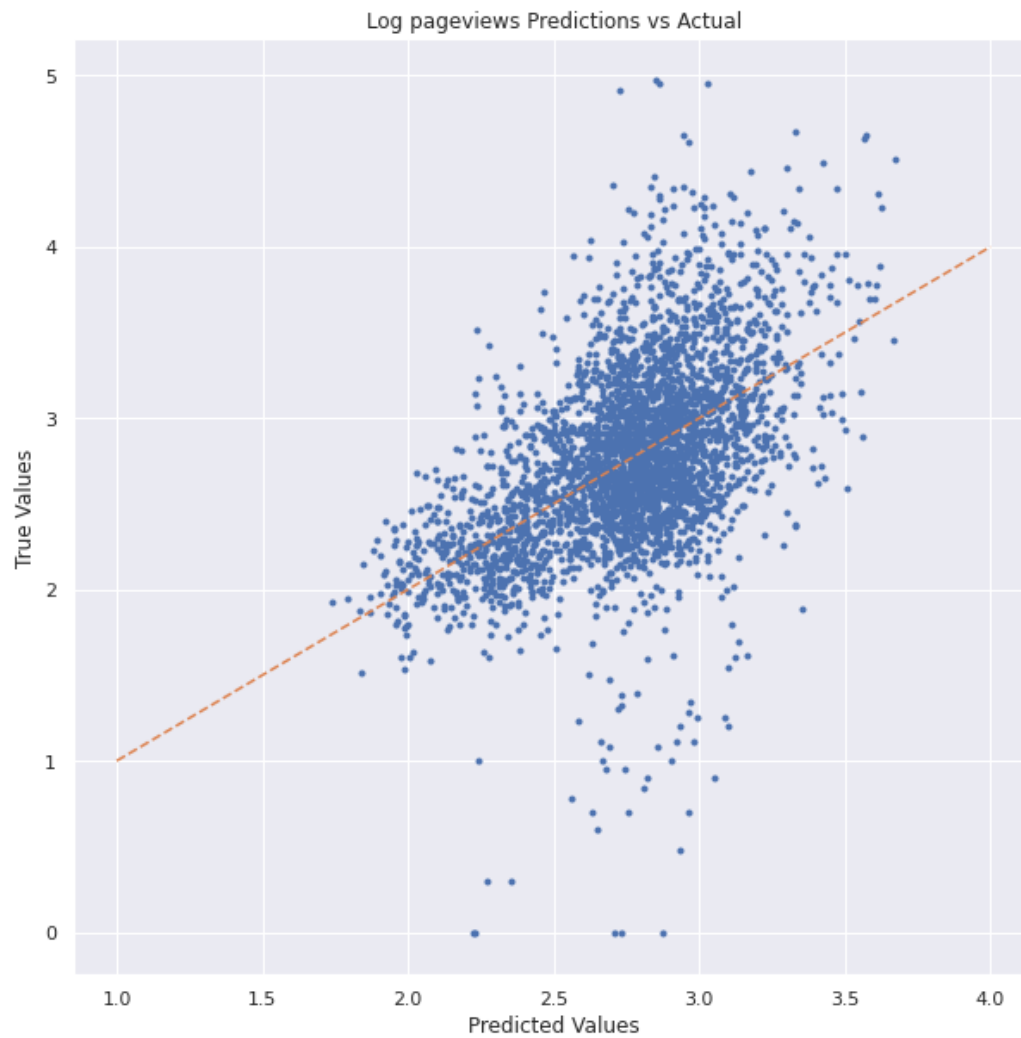
Process

Only non-null articles from 2015 onwards were used for this analysis since this was when engagement started being more accurately tracked. For our predictive variables, everything except time, pageviews and topic proportions (how much an article fits into a given topic) was dropped. This means that we're only going to be predicting engagement based on what topics an article belongs to and when it was published. Additionally, we will be predicting pageviews in log form since this pageviews spans several orders of magnitude. After this process, we will be running and evaluating a linear regression on an 80-20 split of training test data and inspecting the resulting Mean squared error (MSE).

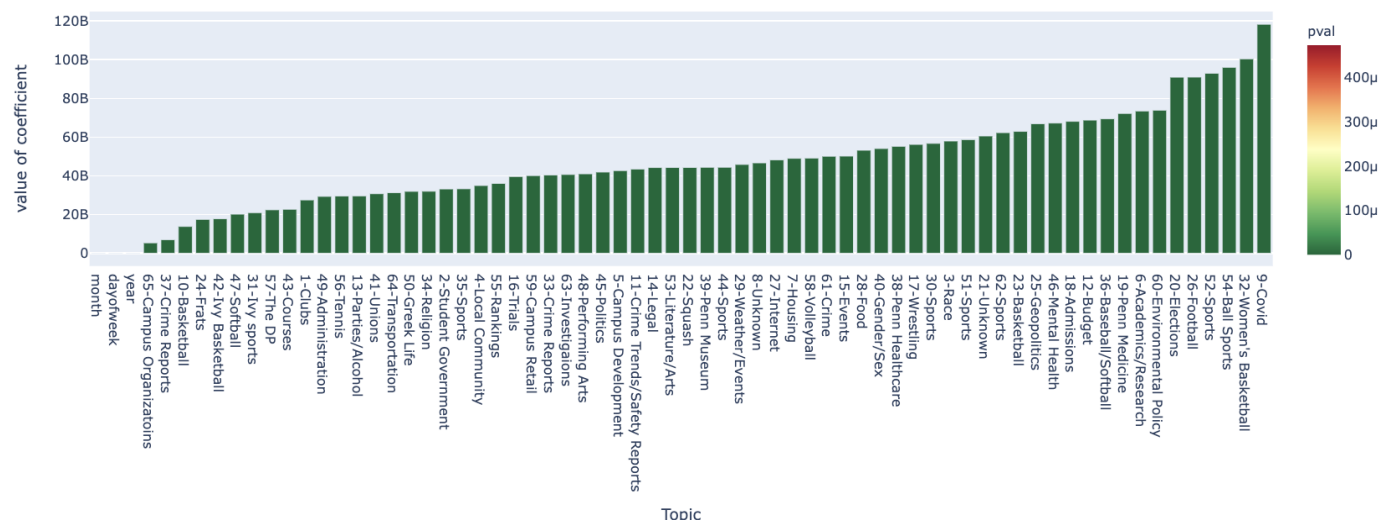
Results

The regression produced a mean squared error of 0.28 on the testing set, which is relatively good but not great. Thinking about the MSE in non-logarithmic terms, it means that our predictions, on average, were overshooting or undershooting the actual pageviews for an article by a factor of two. It predicts 100 views when there were 200 and 200 when there were only actually 100. This sounds initially bad but it makes sense given the range of possible article views it could predict (100 - 10,000)!

The plot below shows the error with performance looking reasonable when comparing the predicted pageviews to the actual pageviews.



Plotting the coefficient values from the regression it's clear that Covid was popular followed by sports. Interestingly this differs from the result of pageviews by topic since we're now relatively less confounded by time as a variable. It's also worth noting that all the coefficients are statistically significant.



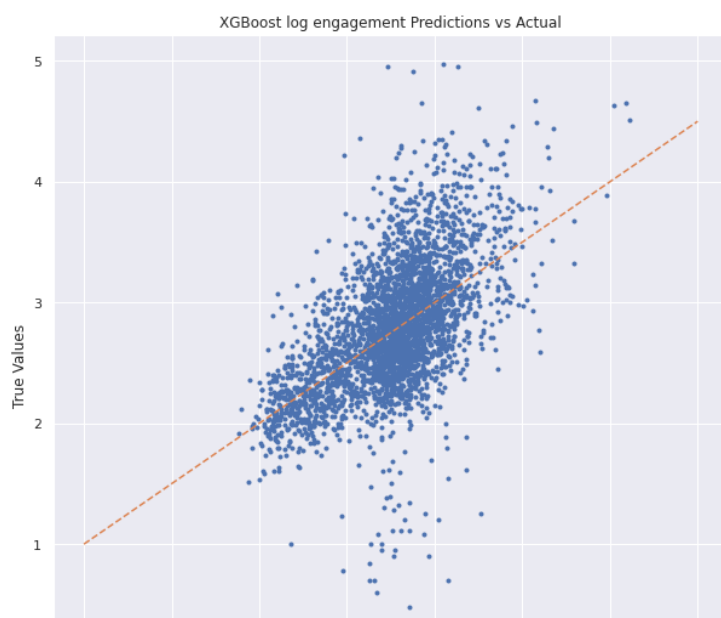
XGBoost Results

To improve our attempt at prediction we used XGBoost since it's one of the best classifiers and provides good results even with minimal tuning. Our XGBoost model below was hand-tuned and passed the same processed data from the linear regression above.

The result was a MSE 0.204, a substantial improvement from the simpler linear regression.

Thinking about this again in non-logarithmic terms, our model, on average is off by a factor of 1.5, which is a considerable improvement from being off, on average, by a factor of two.

The plot below shows the error with performance when comparing the predicted pageviews to the actual pageviews.



Challenges

Data Gathering

Although the dataset was large and clean one limitation was that we gained access to engagement metrics from the DP rather late. This meant it was not included in our project proposal and was also not top-of-mind so it's likely there are more creative uses of it given more time. Additionally, the unreliability of the engagement metrics for articles older than 2015 makes making any very general conclusions difficult.

Data Analysis

Going into the project we expected to glean more from the sentiment analysis than we ultimately did. For example, hot topics like Greek life or mental health were expected to show spikes in sentiment. Instead most topics demonstrated uniform sentiment levels and many topics appeared too noisy to draw any conclusions with the exceptions being race and the environment.

Another limitation on our analysis was that our own judgment and biases were central to the topic modeling since ultimately we had to look at the keyword frequencies in each topic and make calls as to what the topic was. While we took measures to limit this such as individually interpreting the topic and then "averaging" those interpretations there is still a lot of subjectivity in the exercise. Yet that could be said of many areas of data science despite claims of objective analysis and algorithms.

Next Steps

Although we achieved a substantial improvement in our prediction exercise just by shifting to XGBoost it's likely that given more time we can fine tune our predictive capability for article engagement even further.

It might also be of use to check whether our topic interpretations are useful at classifying future DP articles. Given that the DP has an Analytics team of its own this analysis might be of interest to them and could be an opportunity for future collaboration.

This project was the work of Josh Ludan, Amjad Hamza and Andrew Raine.